

Leo Wang

 leowang@terpmail.umd.edu

 667-600-9743

 College Park, Maryland

 leozwang.org

 [linkedin.com/in/leo-z-wang/](https://www.linkedin.com/in/leo-z-wang/)

 github.com/Andallfor

Skills

Languages **Advanced:** C#, Python

Intermediate: TypeScript/JavaScript, HTML/CSS, Java, C++

Technologies Platforms: Unity, NextJS

Libraries: Dear ImGUI, Newtonsoft, OpenCV, Numpy, Selenium, Tensorflow, Matplotlib, TailwindCSS, React, Vite, AWS

Dev. Tools: Visual Studio, Android Studio, Visual Studio Code, Git, Github

Skills ♦ Software Optimization, Large-Scale Data Processing, Data Visualization, Scientific Simulations

Education

2024-27 ♦ **University of Maryland, Bachelor of Science.**

- Double Major: **Computer Science** (Data Science) and **Math** with a **4.0** GPA.
- Advanced Cybersecurity Experience for Students (ACES) Honors College. (Previous)
- Computer Science Departmental Honors Program. (Current)

2024-27 ♦ **UMD Presidential Scholarship** (\$5,000/year)

2024-27 ◇ **National Merit Scholarship** (\$1,000/year)

Experiences

2024-25 ◇ **Cybersecurity Training Platform for MITRE Corporation.**

Web Developer, UMD

- Helped develop a powerful training platform for MITRE Corporation that wrapped the WebGoat application with a custom UI and admin controls for user management.
- Personally built the frontend section of the admin-side pages using NextJS, TailwindCSS and Typescript.

2023-24 ♦ IEEE Aerospace Conference.

Co-Author, Virtual

- **"Enhancing Space Communications: A Novel Approach to Solving the Multi-Satellite Scheduling Problem"**
- Published paper on the novel techniques used in the Mission Visualization Toolkit created during my NASA Internship.
- Accepted into the 2024 IEEE Aerospace Conference.

2021-23 ♦ **National Aeronautics and Space Administration (NASA).**

Intern, Goddard Space Flight Center

- Led a team of up to 5 other interns to create the Mission Visualization Toolkit, a highly-performant, open-source, scientific simulation of planetary terrain and orbiting satellites. The system was built in C# using the Unity Engine.
- As lead developer, I implemented most base systems, project structure, and the planetary terrain system (see below).
- The application specifically focused on analyzing arbitrary lunar case studies and generating highly accurate access calls with Earth satellites by simulating the terrain around ground stations.

Projects

2025- ♦ **Radiology Pronunciation Dictionary for Foreign Speakers.**

NextJS, TailwindCSS, Python

- Commissioned to create a personalized phonetic dictionary to help foreign speakers learn English.
- Internal pronunciation database consists of ~10,000 words, with ~4,000 being radiology terms.
- Implemented functionality to search for words by phoneme and searching for phrases within various radiology textbooks.

2024 ◇ **N-Body (Barnes-Hut) Simulation.**

C++, Dear ImGui

- Developed an optimized N-body simulation (up to 15,000 bodies) by leveraging the Barnes-Hut algorithm.
- Included optimizations such as leapfrog integration, tail recursion, and incremental center of mass calculations.

2022-23 ♦ NASA Mission Visualization Toolkit Terrain System.

C#, Unity

- Designed an extremely powerful and versatile planetary terrain system that served as the backbone of my NASA internship.
- Capable of dynamically converting terrain data into interactable 3D meshes at arbitrary resolutions in realtime.
- Internal database stores highly compressed height maps ranging from 10,000,000 to 1,000,000,000+ points.
- The system can process, render, and simulate up to 13,000,000 points/second (including database read times).